

Supply Chain Analytics — Power BI Model Documentation

File: Supply chain project end to end.pbix **Prepared by:** Isha, Business Intelligence Analyst **Last updated:** August 2026

1. Project Overview

This model gives leadership visibility into supplier performance, delivery bottlenecks, inventory levels, and order fulfillment efficiency across the supply chain — from purchase order placement through delivery to the customer.

Source data: 12 CSV extracts covering Purchase Orders, Purchase Order Lines, Supplier Master, Supplier Performance, Customer Sales Orders, Fulfillment Status, Shipment Tracking, Carriers Master, Warehouse Master, Warehouse Inventory, Stock Movements, and Raw Materials Master.

Primary end users: - **Supply Chain Manager** — full operational visibility across the entire chain - **Executive team** — high-level KPIs and trends for strategic decisions

Model type: Import mode, star schema, single semantic model in Power BI Desktop.

2. Business Problem & Key KPIs

Leadership lacked a unified view connecting four operational areas that were previously siloed:

Business question	Answered by
Are suppliers delivering on time and at quality?	Avg On-Time Delivery Rate , Avg Supplier Quality Score , Avg Supplier Lead Time (Days)
Where are the delivery bottlenecks?	Delayed Shipment Rate , Shipments Delayed , Shipments In Transit
What's our current inventory position?	Total Quantity On Hand , Total Stock Movement Quantity
How efficiently are we fulfilling customer orders?	Order Fulfillment Rate (and its MTD/QTD/YTD/YoY variants)
How much are we spending, and is it trending up or down?	Total Purchase Order Value and its full Time Intelligence family

Headline KPI for this business problem: Delayed Shipment Rate — directly answers “where are the delivery bottlenecks,” which was the most specific pain point in the original ask.

3. Star Schema Design

Design principle

A table is a **fact** if its rows represent an event, transaction, or measurement that accumulates over time and carries numbers to aggregate. It's a **dimension** if its rows are static, descriptive entities used to filter and group.

Dimension tables (5)

Table (current name)	Recommended name	Why it's a dimension
PNRao_SupplyChain_Supplier_Master	Dim_Supplier	Static attributes (name, country) describing a supplier entity
PNRao_SupplyChain_Carriers_Master	Dim_Carrier	Static shipping carrier attributes
PNRao_SupplyChain_Warehouse_Master	Dim_Warehouse	Static warehouse attributes (name, location)
PNRao_SupplyChain_Raw_Materials_Master	Dim_Material	Static material attributes (name, category); StandardUnitCost is a reference attribute, not a transaction
Dim_Date	Dim_Date (already correctly named)	Classic calendar dimension — every fact table's dates roll up through this one shared calendar

Fact tables (8)

Table (current name)	Recommended name	Grain	Why it's a fact
PNRao_SupplyChain_Purchase_Orders	Fact_Purchase_Orders	1 row per PO header	Purchasing event with a measure (TotalCost) and FK to Supplier
PNRao_SupplyChain_Purchase_Order_Lines	Fact_Purchase_Order_Lines	1 row per PO line item	Transactional detail with measures Quantity , UnitCost
PNRao_SupplyChain_Supplier_Performance	Fact_Supplier_Performance	1 row per supplier (periodic KPI snapshot)	Contains measures (delivery rate, quality score) rather than descriptive identity
PNRao_SupplyChain_Customer_Sales_Orders	Fact_Customer_Sales_Orders	1 row per sales order line	Transactional event with Quantity measure
PNRao_SupplyChain_Fulfillment_Status	Fact_Fulfillment_Status	1 row per fulfillment record	Tracks an order's fulfillment event/status
PNRao_SupplyChain_Shipment_Tracking	Fact_Shipment_Tracking	1 row per shipment	Logistics event with FKs to Carrier and Warehouse
PNRao_SupplyChain_Stock_Movements	Fact_Stock_Movements	1 row per inventory movement	Classic transactional "movement fact"
PNRao_SupplyChain_Warehouse_Inventory	Fact_Warehouse_Inventory	1 row per inventory snapshot	Periodic snapshot fact (QuantityOnHand at a point in time)

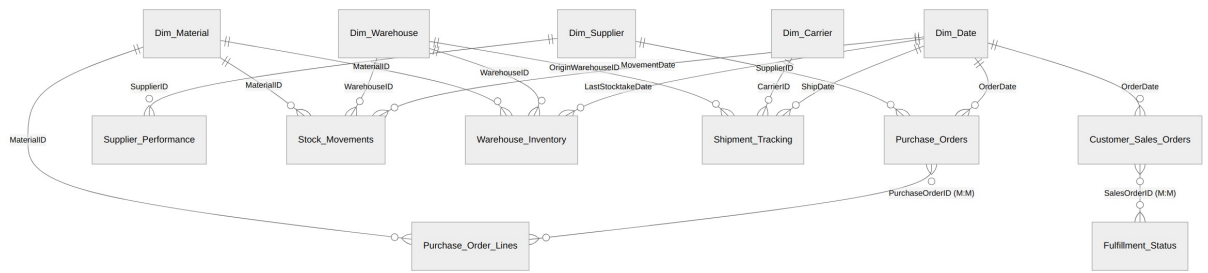
Note: Table renames to the **Dim_ / Fact_** convention were recommended but have not yet been applied in the live file (requires a manual rename in Power BI Desktop's Fields pane — this connection's tooling can edit column/measure/relationship objects but not table-level name or description properties). This document uses current live names throughout; update this table once renamed.

A structural trade-off worth understanding

Several fact tables retain rows flagged **DuplicateFlag = "Duplicate"** — records that share a primary key but have conflicting data. These were **kept, not deleted**, per an explicit decision to preserve all source data for manual review rather than silently discarding potentially-real records. Because of this, two fact-to-fact relationships could not be built as standard One-to-Many (the "one" side isn't actually unique) and are instead **Many-to-Many**: - **Purchase_Order_Lines** ↔ **Purchase_Orders** (via **PurchaseOrderID**) - **Fulfillment_Status** ↔ **Customer_Sales_Orders** (via **SalesOrderID**)

All core business measures filter to **DuplicateFlag = "Unique"** so a conflicting duplicate never silently inflates a total — see the **Data Quality** measure folder for transparency on exactly how many rows are excluded this way.

4. Model Relationship Diagram



Model Relationship Diagram

Full relationship list (19 total)

From table	From column	To table	To column	Cardinality	Cross-filter	Active
Purchase_Orders	SupplierID	Supplier_Master	SupplierID	Many:One	Single	Yes
Supplier_Performance	SupplierID	Supplier_Master	SupplierID	Many:One	Single	Yes
Purchase_Order_Lines	MaterialID	Raw_Materials_Master	MaterialID	Many:One	Single	Yes
Stock_Movements	MaterialID	Raw_Materials_Master	MaterialID	Many:One	Single	Yes
Warehouse_Inventory	MaterialID	Raw_Materials_Master	MaterialID	Many:One	Single	Yes
Stock_Movements	WarehouseID	Warehouse_Master	WarehouseID	Many:One	Single	Yes
Warehouse_Inventory	WarehouseID	Warehouse_Master	WarehouseID	Many:One	Single	Yes
Shipment_Tracking	OriginWarehouseID	Warehouse_Master	WarehouseID	Many:One	Single	Yes
Shipment_Tracking	CarrierID	Carriers_Master	CarrierID	Many:One	Single	Yes
Purchase_Order_Lines	PurchaseOrderID	Purchase_Orders	PurchaseOrderID	Many:Many	Single	Yes
Fulfillment_Status	SalesOrderID	Customer_Sales_Orders	SalesOrderID	Many:Many	Single	Yes
Fulfillment_Status	ShipmentID	Shipment_Tracking	ShipmentID	Many:Many	Single	No (disabled to avoid an ambiguous filter path to Dim_Date; reachable via USERELATIONSHIP)
Purchase_Orders	OrderDate	Dim_Date	Date	Many:One	Single	Yes
Purchase_Orders	ExpectedDeliveryDate	Dim_Date	Date	Many:One	Single	No (role-playing second date on the same fact table)
Customer_Sales_Orders	OrderDate	Dim_Date	Date	Many:One	Single	Yes
Shipment_Tracking	ShipDate	Dim_Date	Date	Many:One	Single	Yes
Stock_Movements	MovementDate	Dim_Date	Date	Many:One	Single	Yes
Warehouse_Inventory	LastStocktakeDate	Dim_Date	Date	Many:One	Single	Yes
Dim_Date	Date	(auto-generated hidden date table)	Date	Many:One	Single	Yes — leftover from Power BI's Auto Date/Time; harmless but unused (see Known Limitations)

5. Data Dictionary

Every table below is documented with its business purpose, current row count, and full column list. Columns marked **[Hidden]** are foreign keys or join-only date columns not intended for direct use — use the linked dimension/ **Dim_Date** table instead.

Dim_Supplier (PNRao_SupplyChain_Supplier_Master) — 3 rows

Master list of suppliers the company sources raw materials from.

Column	Type	Description
SupplierID	String	Unique identifier for a supplier
SupplierName	String	Supplier's business name
Country	String	Country the supplier operates from
DuplicateFlag	String	Data-quality flag: Unique or Duplicate

Dim_Carrier (PNRao_SupplyChain_Carriers_Master) — 4 rows

Master list of shipping carriers used for outbound logistics.

Column	Type	Description
CarrierID	String	Unique identifier for a shipping carrier
CarrierName	String	Shipping carrier's business name
DuplicateFlag	String	Data-quality flag: Unique or Duplicate

Dim_Warehouse (PNRao_SupplyChain_Warehouse_Master) — 3 rows

Master list of warehouse/storage locations.

Column	Type	Description
WarehouseID	String	Unique identifier for a warehouse
WarehouseName	String	Warehouse's name
Location	String	City where the warehouse is located
DuplicateFlag	String	Data-quality flag: Unique or Duplicate

Dim_Material (PNRao_SupplyChain_Raw_Materials_Master) — 4 rows

Master list of raw materials purchased and stocked.

Column	Type	Description
MaterialID	String	Unique identifier for a raw material
MaterialName	String	Raw material's name
Category	String	Product category this material belongs to
StandardUnitCost	Decimal	Reference/standard cost per unit for this material
DuplicateFlag	String	Data-quality flag: Unique or Duplicate

Dim_Date — 1,826 rows (2022-01-01 to 2026-12-31)

Calendar dimension. The single shared date table every fact table filters through.

Column	Type	Description
Date	DateTime	Calendar date — the join key to every fact table
Year	Int64	Calendar year
MonthNo	Int64	Month number (1-12), used to sort Month correctly
Month	String	Month name, e.g. January
Quarter	String	Calendar quarter, e.g. Q1
YearMonth	String	Year and month combined, e.g. 2025-06
DayOfWeek	String	Day of the week name, e.g. Monday
IsWeekend	Boolean	True if Saturday or Sunday
Day	Int64 (calculated)	Day of the month (1-31) — bottom level of the Calendar Hierarchy

Fact_Purchase_Orders (PNRao_SupplyChain_Purchase_Orders) — 11,160 rows

One row per purchase order sent to a supplier.

Column	Type	Description
PurchaseOrderID	String	Unique identifier for a purchase order
SupplierID [Hidden]	String	FK to Dim_Supplier
OrderDate [Hidden]	DateTime	FK to Dim_Date — date the order was placed
ExpectedDeliveryDate [Hidden]	DateTime	FK to Dim_Date (inactive relationship) — date promised by supplier
Status	String	Ordered, Shipped, Delivered, or Cancelled
TotalCost	Decimal	Total dollar value of the entire purchase order
DuplicateFlag	String	Unique, Duplicate, or Missing Key

Fact_Purchase_Order_Lines (PNRao_SupplyChain_Purchase_Order_Lines) — 27,835 rows

One row per line item within a purchase order.

Column	Type	Description
PurchaseOrderLineID	String	Unique identifier for the line item
PurchaseOrderID [Hidden]	String	FK to Fact_Purchase_Orders
MaterialID [Hidden]	String	FK to Dim_Material
Quantity	Int64	Units of material ordered on this line
UnitCost	Decimal	Price paid per unit on this order
DuplicateFlag	String	Unique, Duplicate, or Missing Key

Fact_Supplier_Performance (PNRao_SupplyChain_Supplier_Performance) — 2 rows

Periodic KPI scorecard per supplier.

Column	Type	Description
SupplierID [Hidden]	String	FK to Dim_Supplier
SupplierName	String	Supplier's name as recorded in the scorecard
OnTimeDeliveryRate	Double	Share of this supplier's deliveries that arrived on/before the promised date
QualityScore	Double	Quality rating on the source system's scale
AverageLeadTimeDays	Int64	Average days from order to delivery
DuplicateFlag	String	Unique, Duplicate, or Missing Key

Fact_Customer_Sales_Orders (PNRao_SupplyChain_Customer_Sales_Orders) — 44,586 rows

One row per customer sales order line.

Column	Type	Description
SalesOrderID	String	Unique identifier for a sales order line
OrderDate [Hidden]	DateTime	FK to Dim_Date
ProductID	String	Product identifier (no separate Product dimension exists in this model)
ProductName	String	Name of the product ordered
Quantity	Int64	Units ordered
CustomerID	String	Customer identifier (no separate Customer dimension exists in this model)
DuplicateFlag	String	Unique, Duplicate, or Missing Key

Fact_Fulfillment_Status (PNRao_SupplyChain_Fulfillment_Status) — 44,605 rows

One row per order fulfillment record.

Column	Type	Description
FulfillmentID	String	Unique identifier for the fulfillment record
SalesOrderID [Hidden]	String	FK to Fact_Customer_Sales_Orders
ShipmentID [Hidden]	String	FK to Fact_Shipment_Tracking (inactive relationship)
Status	String	Fulfilled or pending
DuplicateFlag	String	Unique, Duplicate, or Missing Key

Fact_Shipment_Tracking (PNRao_SupplyChain_Shipment_Tracking) — 44,642 rows

One row per shipment.

Column	Type	Description
ShipmentID	String	Unique identifier for a shipment
CarrierID [Hidden]	String	FK to Dim_Carrier
OriginWarehouseID [Hidden]	String	FK to Dim_Warehouse
ShipDate [Hidden]	DateTime	FK to Dim_Date
Status	String	Processing, In Transit, Delayed, or Delivered
DuplicateFlag	String	Unique, Duplicate, or Missing Key

Fact_Stock_Movements (PNRao_SupplyChain_Stock_Movements) — 7,041 rows

One row per inventory movement event.

Column	Type	Description
MovementID	String	Unique identifier for the movement event
MaterialID [Hidden]	String	FK to Dim_Material
WarehouseID [Hidden]	String	FK to Dim_Warehouse
MovementType	String	Direction of movement, e.g. Inbound
Quantity	Int64	Units moved
MovementDate [Hidden]	DateTime	FK to Dim_Date
DuplicateFlag	String	Unique, Duplicate, or Missing Key

Fact_Warehouse_Inventory (PNRao_SupplyChain_Warehouse_Inventory) — 11 rows

Periodic snapshot of inventory on hand.

Column	Type	Description
InventoryID	String	Unique identifier for the inventory record
MaterialID [Hidden]	String	FK to Dim_Material
WarehouseID [Hidden]	String	FK to Dim_Warehouse
QuantityOnHand	Int64	Units currently on hand
LastStocktakeDate [Hidden]	DateTime	FK to Dim_Date — date last physically verified
DuplicateFlag	String	Unique, Duplicate, or Missing Key

_Measures

Dedicated, empty measures table — organizational home for all DAX measures in the model (30 curated measures documented in Section 6). Contains one hidden placeholder column (**Column1**) with no analytical purpose.

6. Measures Catalog (30 total)

Curated to the 30 highest-value measures across all 5 business domains plus a focused Time Intelligence layer. All formulas below are copied directly from the live model. Every core business measure filters to **DuplicateFlag = "Unique"** to exclude conflicting duplicate records from totals.

📦 Purchasing (5)

Total Purchase Orders

```
CALCULATE(COUNTROWS('PNRao_SupplyChain_Purchase_Orders'), 'PNRao_SupplyChain_Purchase_Orders'[DuplicateFlag]="Unique")
```

How many clean, non-conflicting purchase orders exist in the selected period.

Total Purchase Order Value

```
CALCULATE(SUM('PNRao_SupplyChain_Purchase_Orders'[TotalCost]),
'PNRao_SupplyChain_Purchase_Orders'[DuplicateFlag]="Unique")
```

Total dollars committed to suppliers — the core procurement spend number.

Average Purchase Order Value

```
DIVIDE([Total Purchase Order Value], [Total Purchase Orders])
```

Typical size of a single PO — flags whether you're buying in small or large batches.

Open Purchase Orders

```
CALCULATE(COUNTROWS('PNRao_SupplyChain_Purchase_Orders'), 'PNRao_SupplyChain_Purchase_Orders'[Status]="Ordered",  
'PNRao_SupplyChain_Purchase_Orders'[DuplicateFlag]="Unique")
```

Orders placed but not yet shipped/delivered/cancelled — the current purchasing pipeline.

Cancelled Order Rate

```
DIVIDE(CALCULATE(COUNTROWS('PNRao_SupplyChain_Purchase_Orders'), 'PNRao_SupplyChain_Purchase_Orders'[Status]="Cancelled",  
'PNRao_SupplyChain_Purchase_Orders'[DuplicateFlag]="Unique"), [Total Purchase Orders])
```

Share of POs that get cancelled — a proxy for planning/forecasting problems upstream.

Supplier Performance (4)

Avg On-Time Delivery Rate

```
AVERAGE('PNRao_SupplyChain_Supplier_Performance'[OnTimeDeliveryRate])
```

Average reliability of the supplier base — how often they hit promised dates.

Avg Supplier Quality Score

```
AVERAGE('PNRao_SupplyChain_Supplier_Performance'[QualityScore])
```

Average quality rating across suppliers — flags sourcing risk.

Avg Supplier Lead Time (Days)

```
AVERAGE('PNRao_SupplyChain_Supplier_Performance'[AverageLeadTimeDays])
```

How long, on average, suppliers take to deliver — feeds reorder-point planning.

Total Active Suppliers

```
DISTINCTCOUNT('PNRao_SupplyChain_Supplier_Master'[SupplierID])
```

Size of the supplier base — relevant for concentration/dependency risk.

Sales & Fulfillment (5)

Total Sales Orders

```
CALCULATE(COUNTROWS('PNRao_SupplyChain_Customer_Sales_Orders'), 'PNRao_SupplyChain_Customer_Sales_Orders'[DuplicateFlag]  
="Unique")
```

Demand volume — how many customer orders came in.

Total Units Sold

```
CALCULATE(SUM('PNRao_SupplyChain_Customer_Sales_Orders'[Quantity]), 'PNRao_SupplyChain_Customer_Sales_Orders'[Duplicate  
Flag]="Unique")
```

Total product volume shipped to customers.

Distinct Customers

```
CALCULATE(DISTINCTCOUNT('PNRao_SupplyChain_Customer_Sales_Orders'[CustomerID]), 'PNRao_SupplyChain_Customer_Sales_Order  
s'[DuplicateFlag]="Unique")
```

Breadth of the customer base in the period.

Fulfilled Orders

```
CALCULATE(COUNTROWS('PNRao_SupplyChain_Fulfillment_Status'), 'PNRao_SupplyChain_Fulfillment_Status'[Status]="Fulfilled",  
'PNRao_SupplyChain_Fulfillment_Status'[DuplicateFlag]="Unique")
```


Count of orders successfully completed end-to-end.

Order Fulfillment Rate

```
DIVIDE([Fulfilled Orders],  
CALCULATE(COUNTROWS('PNRao_SupplyChain_Fulfillment_Status'),'PNRao_SupplyChain_Fulfillment_Status'[DuplicateFlag]="Unique"))
```

The single most important ops KPI — what % of orders actually get fulfilled.

📦 Logistics & Shipping (5)

Total Shipments

```
CALCULATE(COUNTROWS('PNRao_SupplyChain_Shipment_Tracking'),'PNRao_SupplyChain_Shipment_Tracking'[DuplicateFlag]="Unique")
```

Total shipping activity in the period.

Shipments In Transit

```
CALCULATE(COUNTROWS('PNRao_SupplyChain_Shipment_Tracking'),'PNRao_SupplyChain_Shipment_Tracking'[Status]="In Transit", 'PNRao_SupplyChain_Shipment_Tracking'[DuplicateFlag]="Unique")
```

Shipments currently moving — visibility into what's "in flight" right now.

Shipments Delayed

```
CALCULATE(COUNTROWS('PNRao_SupplyChain_Shipment_Tracking'),'PNRao_SupplyChain_Shipment_Tracking'[Status]="Delayed", 'PNRao_SupplyChain_Shipment_Tracking'[DuplicateFlag]="Unique")
```

Count of shipments running behind — the direct signal for delivery-bottleneck reporting.

Delayed Shipment Rate

```
DIVIDE([Shipments Delayed],[Total Shipments])
```

What share of all shipments are delayed — the headline logistics-health KPI.

Shipments Delivered

```
CALCULATE(COUNTROWS('PNRao_SupplyChain_Shipment_Tracking'),'PNRao_SupplyChain_Shipment_Tracking'[Status]="Delivered", 'PNRao_SupplyChain_Shipment_Tracking'[DuplicateFlag]="Unique")
```

Completed shipments — the throughput number.

📦 Inventory & Warehouse (3)

Total Quantity On Hand

```
CALCULATE(SUM('PNRao_SupplyChain_Warehouse_Inventory'[QuantityOnHand]), 'PNRao_SupplyChain_Warehouse_Inventory'[DuplicateFlag]="Unique")
```

Current inventory position across all warehouses — the core stock-level number.

Total Stock Movement Quantity

```
CALCULATE(SUM('PNRao_SupplyChain_Stock_Movements'[Quantity]), 'PNRao_SupplyChain_Stock_Movements'[DuplicateFlag]="Unique")
```

Total volume of goods moving in/out of warehouses in the period.

Total Warehouses

```
DISTINCTCOUNT('PNRao_SupplyChain_Warehouse_Master'[WarehouseID])
```

Number of active storage locations.

Time Intelligence Purchase Order Value (3)

Total Purchase Order Value YTD

```
TOTALYTD([Total Purchase Order Value], 'Dim_Date'[Date])
```

Purchase order value so far this year.

Total Purchase Order Value YoY Growth %

```
VAR Curr = [Total Purchase Order Value]
VAR Prior = CALCULATE([Total Purchase Order Value], SAMEPERIODLASTYEAR('Dim_Date'[Date]))
RETURN DIVIDE(Curr - Prior, Prior)
```

Percentage change in spend versus the same period last year.

Total Purchase Order Value Rolling 3M Avg

```
VAR CurrentDate = MAX('Dim_Date'[Date])
VAR Last3M = DATESINPERIOD('Dim_Date'[Date], CurrentDate, -3, MONTH)
RETURN DIVIDE(CALCULATE([Total Purchase Order Value], Last3M), 3)
```

Average monthly spend over the trailing 3 months — smooths out month-to-month noise.

Time Intelligence Sales Orders (3)

Total Sales Orders YTD

```
TOTALYTD([Total Sales Orders], 'Dim_Date'[Date])
```

Sales orders so far this year.

Total Sales Orders YoY Growth %

```
VAR Curr = [Total Sales Orders]
VAR Prior = CALCULATE([Total Sales Orders], SAMEPERIODLASTYEAR('Dim_Date'[Date]))
RETURN DIVIDE(Curr - Prior, Prior)
```

Percentage change in order volume versus the same period last year.

Total Sales Orders Rolling 3M Avg

```
VAR CurrentDate = MAX('Dim_Date'[Date])
VAR Last3M = DATESINPERIOD('Dim_Date'[Date], CurrentDate, -3, MONTH)
RETURN DIVIDE(CALCULATE([Total Sales Orders], Last3M), 3)
```

Average monthly order volume over the trailing 3 months.

Time Intelligence Fulfillment Rate (3)

Order Fulfillment Rate YTD

```
TOTALYTD([Order Fulfillment Rate], 'Dim_Date'[Date])
```

Fulfillment rate calculated across the current year-to-date.

Order Fulfillment Rate YoY Growth %

```
VAR Curr = [Order Fulfillment Rate]
VAR Prior = CALCULATE([Order Fulfillment Rate], SAMEPERIODLASTYEAR('Dim_Date'[Date]))
RETURN Curr - Prior
```

Percentage-point change versus last year (not percent-of-a-percent) — “the rate improved 3.4 points” is the intended reading.

Order Fulfillment Rate Rolling 3M Avg

```
VAR CurrentDate = MAX('Dim_Date'[Date])
VAR Last3M = DATESINPERIOD('Dim_Date'[Date], CurrentDate, -3, MONTH)
RETURN CALCULATE([Order Fulfillment Rate], Last3M)
```

The **blended** fulfillment rate over the trailing 3 months (total fulfilled ÷ total orders across all 3 months) — deliberately not a simple average of 3 monthly percentages, which would misweight months with different order volumes.

Note: The live Power BI model also contains MTD, QTD, SPLY, and MoM Growth % variants for each of these 3 metrics, plus a dedicated Data Quality folder tracking flagged duplicate counts — 50 measures in total. This document highlights the 30 most business-relevant for review purposes.

7. Hierarchies

Four drill-down hierarchies were built for report navigation:

Hierarchy	Table	Levels
Calendar Hierarchy	Dim_Date	Year → Quarter → Month → Day
Supplier Hierarchy	Dim_Supplier	Country → Supplier
Warehouse Hierarchy	Dim_Warehouse	Location → Warehouse
Material Hierarchy	Dim_Material	Category → Material

Month is sorted by the numeric **MonthNo** column so drill-downs display in calendar order (January → December), not alphabetically.

8. Row-Level Security (RLS)

Three roles restrict data access by job function. Because all fact-table relationships filter single-direction from dimension → fact, a filter placed on a dimension table automatically cascades to every related fact table — no need to filter each fact table individually.

Role	Model Permission	Filter	Effect
Supply Chain Manager	Read	(none)	Full, unrestricted access to every table
Supplier View	Read	<code>Dim_Supplier[SupplierName] = USERPRINCIPALNAME()</code>	Sees only their own supplier's row in Dim_Supplier — cascades to Purchase_Orders and Supplier_Performance
Warehouse Manager	Read	<code>Dim_Warehouse[WarehouseName] = USERPRINCIPALNAME()</code>	Sees only their own warehouse's row in Dim_Warehouse — cascades to Stock_Movements, Warehouse_Inventory, and Shipment_Tracking

Important caveat: `USERPRINCIPALNAME()` returns the logged-in user's email/UPN in the Power BI Service. The filters above match it directly against `SupplierName` / `WarehouseName`, which only works if each supplier or warehouse contact's login UPN is provisioned to exactly match their company/location name — unlikely in a real deployment. Before going live, add an `Email` column to `Dim_Supplier` / `Dim_Warehouse` (or a separate user-mapping table) and update the filter to `[Email] = USERPRINCIPALNAME()`.

Note: A fourth role, **Executive**, exists in the file but was built independently, outside this documented workflow. It originally had a `TRUE()` filter on `_Measures`, which looked like a restriction but was actually a no-op (always evaluates true, so it restricted nothing) — that filter permission has since been removed entirely, so Executive now has clean, unrestricted full access exactly like Supply Chain Manager.

9. Data Quality Framework

Every fact table carries a `DuplicateFlag` column with three possible values:

- **Unique** — clean record, included in all core measures
- **Duplicate** — shares a primary key with another row but has conflicting data; **kept, not deleted**, and excluded from core measures by design; counted separately in the Data Quality measure folder
- **Missing Key** — had a null/placeholder primary key; these rows were removed entirely during cleanup (unlike Duplicates, a missing key carries no useful information to preserve)

Additional cleanup applied across all source tables: leading/trailing whitespace trimmed, `"#N/A"` text and blank strings converted to true nulls, and text/date/currency data types corrected to match actual content.